

STAT 801A

Statistical Methods In Research: Non Calculus

Fall 2026

Instructor

Name: Tyler Wiederich, Ph.D.

Office: Hardin Hall XXX

Office hours: TBD.

E-mail: twiederich2@unl.edu

Lab Instructor: XXX

Office: XXX

Office hours XXX

E-mail: XXX

All office hours are guaranteed for the first 30 minutes. If no students are present after 30 minutes, office hours are ended.

Test

Course Description

Prerequisites: Introductory course in statistics.

Statistical concepts and statistical methodology useful in descriptive, experimental, and analytical study of biological and other natural phenomena. Practical application of statistics rather than on statistical theory.

Statistical Software

The statistical computing software package R will be used extensively to perform calculations in this class. Additionally, we will make use of RStudio (GUI for R) and Quarto (document creation).

1. Download R: <https://cran.r-project.org>
2. Download RStudio: <https://posit.co/downloads>
3. Download Quarto: <https://quarto.org/docs/download/>

Assessments

This course consists of quizzes, projects, and exams.

Quizzes: Weekly quizzes will be given in Canvas. Unless otherwise stated, they are due by 9:45 a.m. on Tuesdays.

Projects: Four group projects (2-3 people/group) will be assigned throughout the semester.

Exams: There will be 2 exams during the semester. Each exam will have a closed-book and open-book portion. More details will be provided in class.

Final: To be announced in class

These assessments are weighted as follows:

	Quizzes	Projects*	Exams**	Final
% of grade	10%	40%	30%	20%

*Each project is worth 10% of your grade.

**Each exam is worth 15% of your grade.

All projects need to be turned electronically via PDF documents. A project completed in an unreadable or unprofessional manner will be returned for a zero grade. No late projects or quizzes are accepted.

For group projects, all group members are expected to participate equally and have a complete understanding of all components for it. I will lower a student's project grade if he/she/they do not abide by this group work policy.

Grades

Your overall course grade percentage guarantees the following grade for the course.

Letter Grade	+	-
A	93	90
B	87	83
C	77	73
D	67	63
F	<60	

Continuity of Instruction Policy

In the event that the university is closed during our regular class hours, please check the Canvas announcements for alternative class arrangements.

AI Disclaimer

The recent popularity of large language models and other artificial intelligence tools introduces a unique challenge in the academic community. As an instructor, my goal is to promote critical thinking for the topics taught in this course. Unless otherwise stated, **the use of AI tools is not permitted**. If you are suspected of using AI, I reserve the right to give a zero grade.

Tentative Schedule

Week	Monday	Topics	Notes
1	8/24	Intro to R	No lab
2	8/31	Data summaries	
3	9/7	Probability	No lab (Labor Day)
4	9/14	Probability	
5	9/21	Inference for one mean	
6	9/28	Inference for two means	
7	10/5	Review + Exam 1	
8	10/12	Inference for proportions	
9	10/19	Inference for Variances	No class on Tuesday (Fall break)
10	10/26	ANOVA	
11	11/2	ANOVA	No class on Tuesday – Go Vote!
12	11/9	Review + Exam 2	
13	11/16	Regression	
14	11/23	No class/lab	Thanksgiving
15	11/30	Regression	
16	12/7	Dead week	
17	12/14	Final*	

*The final will be held in-person on Monday, Dec. 14 from 10:00 to noon.

UNL Course Policies and Resources

Students are responsible for knowing the university policies and resources found on this page:

<https://go.unl.edu/coursepolicies>

- University-wide Attendance Policy
- Academic Honesty Policy
- Services for Students with Disabilities
- Mental Health and Well-Being Resources
- Final Exam Schedule
- Fifteenth Week Policy
- Emergency Procedures
- Diversity & Inclusiveness
- Title IX Policy
- Other Relevant University-Wide Policies